

Phrase 3

Table 1. Classification Report

Class	Precision	Recall	F1-Score	Support
FAKE	0.89	0.40	0.55	10,000
REAL	0.61	0.95	0.74	10,000
Accuracy	—	—	0.67	20,000
Macro Average	0.75	0.67	0.65	20,000
Weighted Average	0.75	0.67	0.65	20,000

Table 2. Baseline Evaluation Metrics

Metric	Value
Accuracy	0.6748
Precision	0.6131
Recall	0.9481
F1-Score	0.7446
Specificity	0.4016
Sensitivity	0.9481
AUC-ROC	0.8278

Table 3. Confusion Matrix

	Predicted FAKE	Predicted REAL
Actual FAKE	9,481 (TP)	519 (FN)
Actual REAL	5,984 (FP)	4,016 (TN)

Table 4. Scientific Interpretation

Observation	Interpretation
High Recall (0.9481)	The model correctly identifies most positive samples.
Low Specificity (0.4016)	Many real images are incorrectly classified, indicating poor discrimination of the negative class.
Overall Behavior	The baseline model is biased toward predicting the positive class.
Key Finding	Spatial features extracted from 32×32 source images are insufficient for balanced deepfake detection.
Motivation	These results justify incorporating frequency-domain (DCT) features to improve classification performance.



